MATH317 · NUMERICAL METHODS

Theory Review

All the theory in one file · true/false and definitions

Muharib

What's inside

01 True / false of midterm 472

02 Chapter 1.1: errors and rounding

03 Chapter 1.2: Taylor

04 Chapter 2: floating-point

05 Chapter 3: bisection and Newton

06 Check yourself: new true / false

01 True / false of midterm 472

Statement	Answer	Reason
Significant digits are only important when rounding final results, not during intermediate calculations.	**F**	They matter at every step. Rounding midway grows the error.
Horner's algorithm is used for efficiently evaluating polynomials.	**T**	
The Taylor series converges rapidly for points far from the expansion point.	**F**	Rapidly near the point, not far from it.
The Bisection method guarantees finding the exact root in a finite number of steps.	**F**	It gives an approximation, not the exact root.
Rounding replaces a number with an approximation that minimizes error, whereas chopping discards digits without adjustment.	**T**	
The Mean-Value Theorem is a special case of Taylor's Theorem with zero derivatives.	**F**	It is the case `n = 0`, not "zero derivatives".
The error in the Bisection method after n steps is at most (b – a) divided by 2^n.	**F**	The correct power is `2^{n+1}`.

02 Chapter 1.1: errors and rounding

Term	Definition
Significant digits	Digits counted from the first nonzero digit on the left. 0.0035 has 2; 30.16 has 4.
Absolute error	$\lvert \alpha - \beta \rvert$
Relative error	$\frac{\lvert \alpha - \beta \rvert}{\lvert \alpha \rvert}$ (divide by the exact value)
Accurate to n decimal places	n trusted digits after the decimal point
Accurate to n significant digits	n trusted digits from the first nonzero digit
Chopping	Drop the extra digits without adjustment
Rounding	The nearest n-digit number, so the error is smaller
Horner's algorithm	Evaluates a polynomial without powers. Also called nested multiplication or synthetic division.
Deflation	If Horner's last number is zero, r is a root and the rest are the coefficients of a polynomial one degree lower.

A result is never more accurate than the least accurate number used.
The relative error is the better measure of accuracy.
Errors are measured to judge accuracy and to decide when an iterative method should stop (stopping criteria).

03 Chapter 1.2: Taylor

Point	What to remember
Maclaurin series	Taylor about $c = 0$
Convergence speed	Fast **near** the expansion point; slow or none far from it
Polynomial	The series stops after finitely many terms and is exact
Truncation	Stopping the series after some terms; the difference is the remainder or error term
Remainder	$E_{n+1} = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot (x - c)^{n+1}$ with ξ between c and x
In terms of h	$E_{n+1} = O(h^{n+1})$
When does it represent f?	When the remainder goes to zero. e^x for every x; $\ln(1+x)$ only for $0 \le x \le 1$
Mean-Value Theorem	The case $n = 0$ of Taylor: $f(b) - f(a) = (b - a)\, f'(\xi)$
Alternating series	If the terms decrease to zero it converges, and the error ≤ the first omitted term

04 Chapter 2: floating-point

Term	Definition
Machine numbers	Numbers stored exactly. Finitely many; most reals are not among them (e.g. 0.1)
Overflow	Larger than the largest storable value
Underflow	Closer to zero than the smallest storable value
Hole at zero	The gap between zero and the smallest positive stored number
Machine epsilon	Smallest ε with $1 + \varepsilon \neq 1$. Single: 2^{-23}; double: 2^{-52}
Roundoff error	The difference from the nearest machine number: $fl(x) = x(1 + \delta)$, $\lvert\delta\rvert \leq u$
Unit roundoff	Largest relative error of one rounding. Single 2^{-24}; double 2^{-53}
Loss of significance	Subtracting nearly equal numbers destroys most correct digits

	Single	Double
Bits	32 = 1 + 8 + 23	64 = 1 + 11 + 52
Bias	127	1023
Largest	$\approx 3.4 \times 10^{38}$	$\approx 1.8 \times 10^{308}$
Hex digits	8	16

The leading 1 of the normalized form (1.f) is not stored in the fraction field.

05 Chapter 3: bisection and Newton

Point	What to remember
Root	`f(r) = 0`
Intermediate Value Theorem	f continuous and `f(a)·f(b) < 0` → at least one root between a and b
Sign change	Sufficient, not necessary: if `f(a)·f(b) > 0` there may or may not be a root
Bisection	Also the binary-search method. Reliable but slow; never the exact root. Error $\leq \frac{b-a}{2^{n+1}}$
False Position	Like bisection, but uses where the chord between the ends crosses the x-axis
Newton	Also Newton-Raphson. Needs a differentiable function. Very fast (quadratic convergence: correct digits roughly double), but convergence is guaranteed only if x_0 is close to the root
Origin of Newton's formula	The root of the tangent line, or the first two Taylor terms
Stopping	When $\lvert\varepsilon_a\rvert$ drops below the tolerance, or the iterations exceed n_max

06 Check yourself: new true / false

1. The relative error is computed by dividing by the approximate value.
2. Chopping 0.475 to two digits gives 0.48.
3. A Maclaurin series is a Taylor series about c = 0.
4. The Taylor series of a polynomial is exact after a finite number of terms.
5. The number 0.1 can be stored exactly in binary.
6. Single precision uses a bias of 1023.
7. Underflow happens when a number is too large to store.
8. Subtracting two nearly equal numbers can cause loss of significance.
9. If $f(a) \cdot f(b) > 0$ there is no root in [a, b].
10. Newton's method needs the derivative of the function.
11. Newton's method always converges for any starting point.
12. In single precision the leading 1 of 1.f is stored in the fraction field.

#	Answer	Reason
1	**F**	Divide by the exact value.
2	**F**	Chopping gives 0.47; 0.48 is rounding.
3	**T**	
4	**T**	
5	**F**	It never ends in binary: 0.000110011...
6	**F**	127. 1023 is for double.
7	**F**	That is overflow.
8	**T**	
9	**F**	There may or may not be a root.
10	**T**	
11	**F**	x_0 must be close to the root.
12	**F**	It is not stored; only the bits after the point are.